

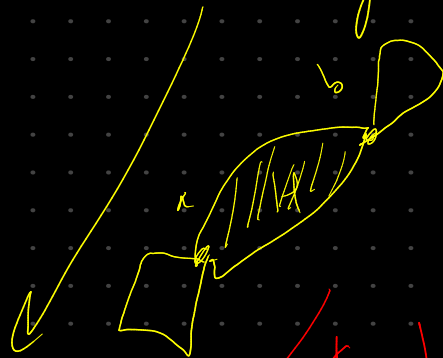
Theorem 2:

h has an π -monotone drawing in which every pair of independent edges crosses evenly

$\Rightarrow$ h has an π -monotone embedding in which each vertex keeps its π -coordinate

a minimal counter-example of TH2 needs to have

h is connected



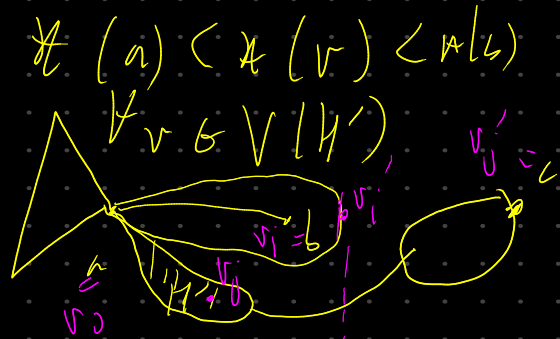
h has no $H \subset h$ s.t. $a, b \in V(H) \setminus V(H)$ s.t. $\pi(a) < \pi(v) < \pi(b)$ $\forall v \in V(H)$, $N(H) = \{a, b\}$ and $V(h) \setminus (V(H) \cup \{a, b\}) \neq \emptyset$



not allowed in min. counterexample

if h has a cut-vertex a and $h - \{a\}$ has a comp H s.t. $\pi(a) < \pi(v) \forall v \in V(H)$

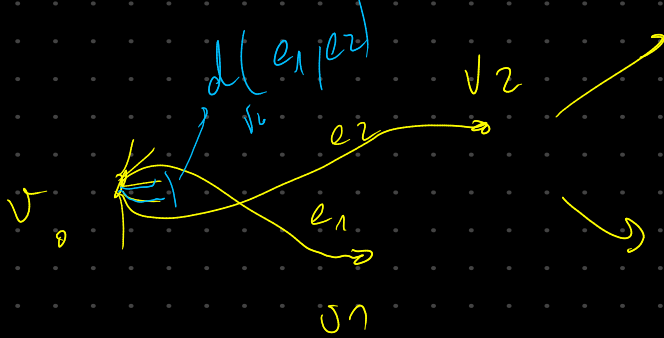
$\Rightarrow V(H) = \{b\}$ $\neq \emptyset$ with $\pi(a) < \pi(v) < \pi(b)$ $\forall v \in V(H')$



we know that ind. edges have even crossing
 what's left to prove is that adjacent edges
 don't affect the planarity, the idea would
 be to shape the curves so we have
 even crossing for them, if that's not possible
 what can we do?

* we start the proof by the init. of the induction
 step

← one odd pair

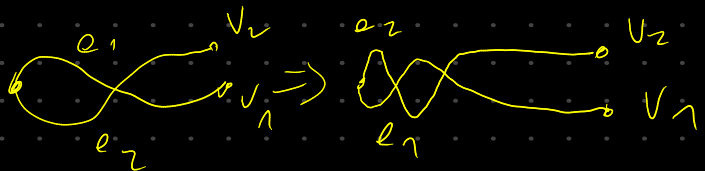


make it cross evenly

we are in a situation
 that has been excluded
 by lemma 3

let h be a minimal counterexample to the theorem.
 we fix a Γ of h which minimizes the number
 of odd pairs.

does changing the rotation affects the
 other crossing (it will affect the
 edges that also have v_0 as a common vertex.
 if $\text{wind}(e_1, e_2) = 0$ then we can switch their
 pos. in $\text{rot}(v_0)$



then they won't be considered

what's left are the edges e_1, e_2 that

$$\min d_{v_0}(e_1, e_2) > 0 \Rightarrow \exists e_3 \text{ rot}_{v_0}(e_1) < \text{rot}_{v_0}(e_3) < \text{rot}_{v_0}(e_2)$$

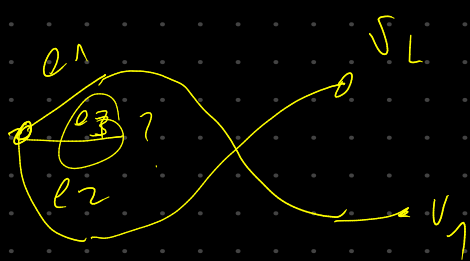
minimal counter-example with min odd crossing

↓
remove one vertex no odd crossing

meaning that the edges incident to
the one vertex (we call it v_0) have odd
crossings.

if we prove that they can have even crossing
the min-counter. is contradicted

the case that is left:

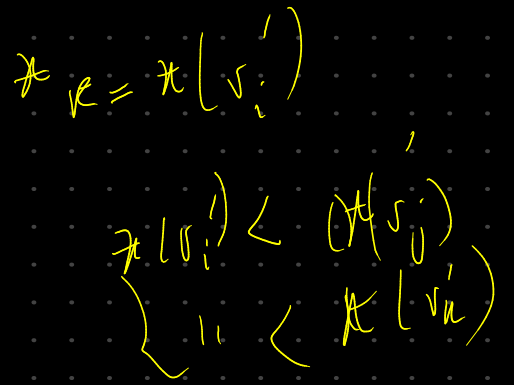


e_3 even # of crossings. $\nearrow$ or $\searrow$ e_2
due to minimality.

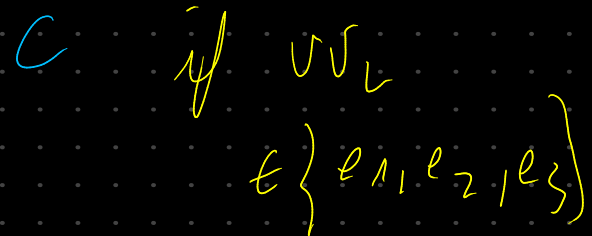
$h_0 \subset h$ with $\forall v \in V(h_0) \quad \kappa(v) \geq \kappa(v_0)$

~~Case 1:~~ v_1, v_2, v_3 are in different components,
of $h_0 - v_0$

No has
 no edges
 (u, v)
 with
 $x(u) \leq x(v)$
 so cut-vertex



✗ Not Case 1. Two of v_1, v_2, v_3 are in the same component C if v_1, v_2 are in the same component.



$$\{i, j, k\} = \{1, 2, 3\} \quad v_i \neq v_L$$

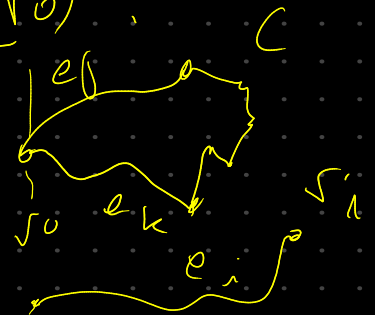
By lemma 5 $\kappa(v_i) < \kappa(v_L) \Rightarrow v_i \in V(C) - v_L$

Let's go back to Lemma 5:

suppose for distinct $j, k \in \{1, 2, 3\}$,
 $\exists$ cycle C that contains e_j and e_k
 st. $\forall v \in V(C) \quad \kappa(v) \geq \kappa(v_0)$

let $v_L \in V(C)$

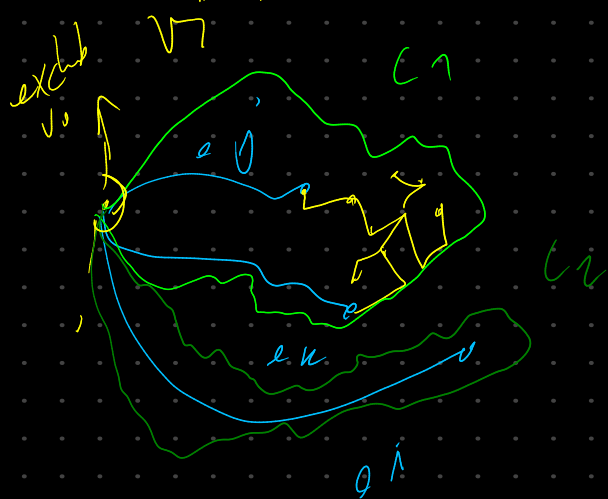
will $\max_{v \in V(C)} (\kappa(v))$



let i be unique index st $\{i, j, k\} = \{1, 2, 3\}$
 suppose $v_i \notin V(C)$

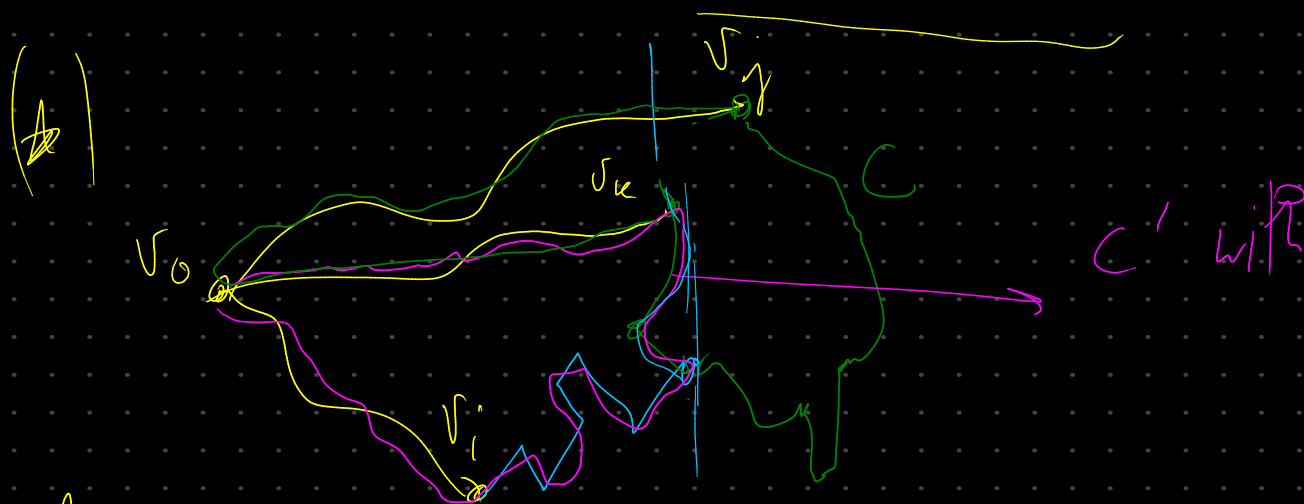
let h'_i be the comp. of $h - V(C)$ that contains v_i

$\forall v \in h'_i \quad \kappa(v_0) < \kappa(v) < \kappa(v_i)$



with lemma 5 $\rightarrow h(v_i) < h(v_L)$

$\downarrow v_i \in V(C) - v_L$



choice of
the smallest
subgraph s.t.
it has a
component that
contains at least
half of v_i, v_k

$P_{i,k}$ should
have
been v_L

contradiction $\rightarrow v_i \notin V(C) - v_L$

$$h(v_i) < h(v_L)$$

or we can
mention that

$h(v_k) < h(v_L)$ from C'
also in this case which
contradicts.

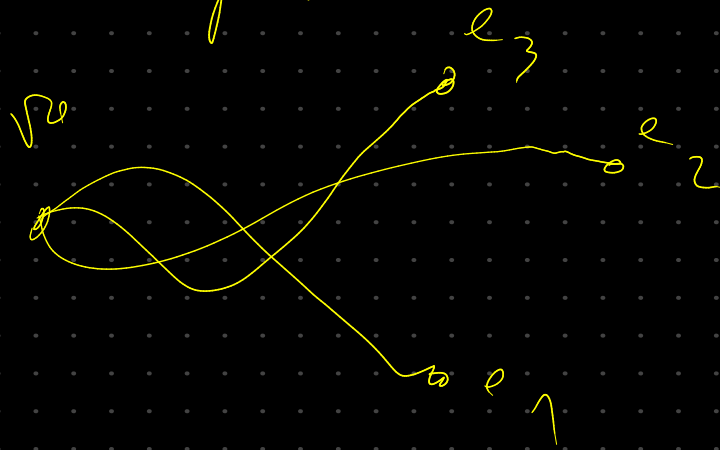
let h'_i be the component in $h - V(C)$ that
 $v_i \in h'_i$

$$v_i \in h_i'$$

$\Rightarrow v_i$ has no neighbors in $V(L) \setminus \{v_0, v_L\}$

because we can't connect v_i for the cycle C if we don't consider v , what about v_L too?

So we go back to the case 1,

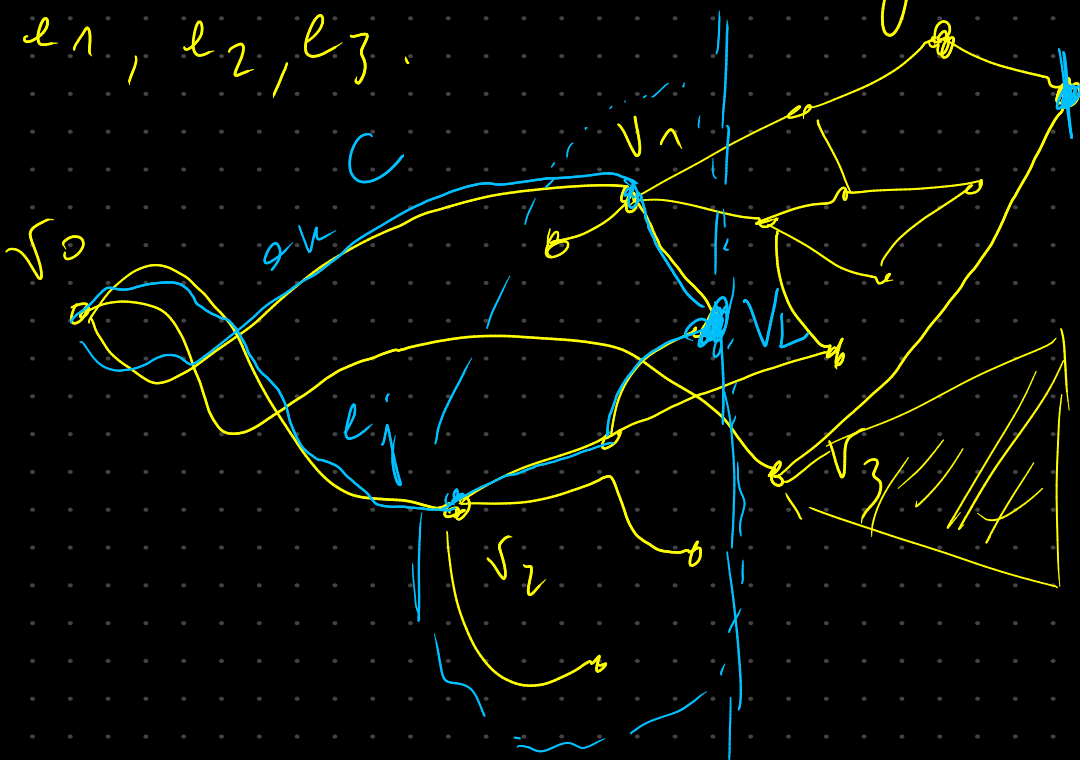


case 1;
distinct,
 u_1, u_2, u_3
in $h_0 - v_0$

case 2
 u_i is not adjacent
to v_L

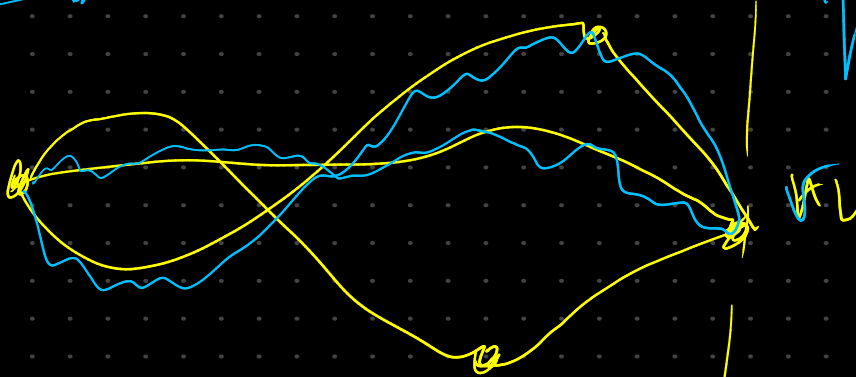
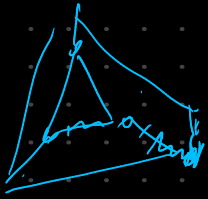
retryp: case not 1

let v_L with $\kappa(v_L)$ the smallest so that
the subgraph induced by vertices v such that
 $\kappa(v_0) < \kappa(v) \leq \kappa(v_L)$ has a component
that contains at least two right endpoints
of e_1, e_2, e_3 .



Let i be unique in $\{i, j, k\} = \{1, 2, 3\}$.

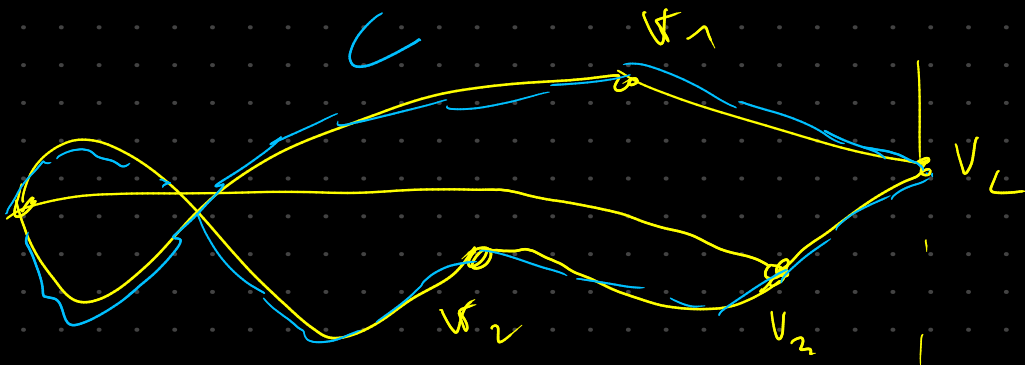
so $v_i \neq v_L$ because if it was



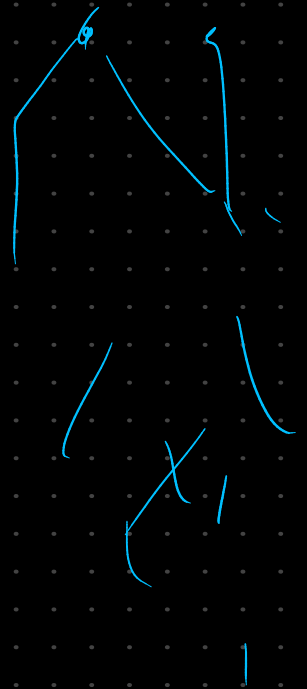
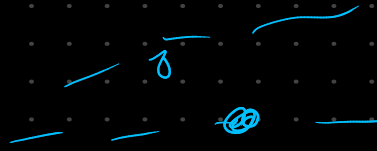
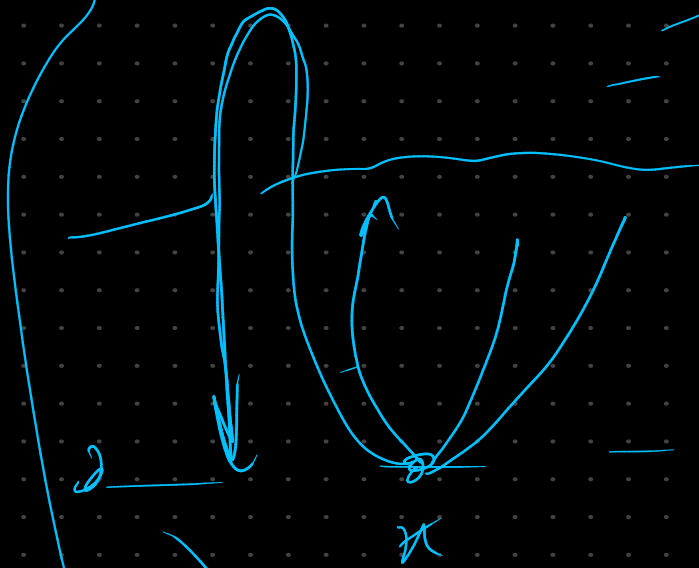
$v_j \neq v_k$
 $\Rightarrow v_i \neq v_L$

By lemma 5 $\kappa(v_i) < \kappa(v_L)$ or $v_i \in V(L) -$

suppose that $\exists Q$ from v_i to $C \equiv v_L$
 $\kappa(v_0) < \kappa(v_1) < \kappa(v_L)$ $\left[\kappa(v_i) < \kappa(v_L) \right]$
 from lemma 5

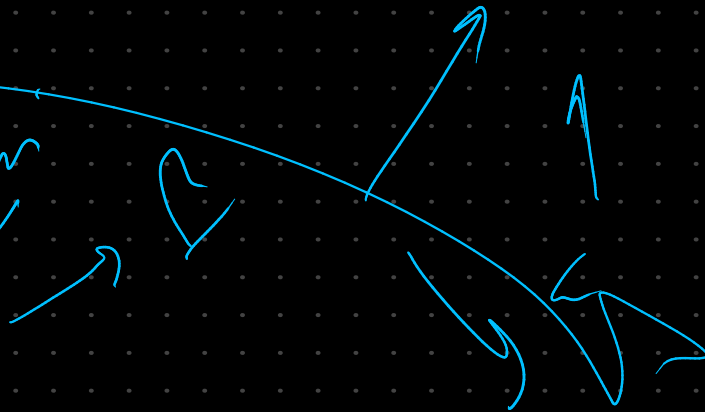


Manan-Tatte



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~~test plan~~



test plan

